Week 3 – Spatial Description and Transformation

ELEC0129 Introduction to Robotics

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- Inverse of Homogeneous Transformation Matrix
- More on Representation of Orientation
- Tutorial Questions

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Inverse of Transformation Matrix (1)

- Given ${}^{A}_{B}T = \left[\begin{array}{ccc|c} & {}^{A}_{B}R & & {}^{A}P_{Borg} \\ \hline 0 & 0 & 0 & 1 \end{array}\right]$,
- How do we get its inverse i.e. ${}^{B}_{A}T$ or ${}^{A}_{B}T^{-1}$?

- We can of course calculate the values using determinants and adjuncts, or by using software.
- However, this does not make use of information which we already know from ${}^{A}_{B}T$!

Inverse of Transformation Matrix (2)

- Let's think this through.
- We know ${}^{A}_{B}T = \left[\begin{array}{ccc|c} & {}^{A}_{B}R & & {}^{A}P_{Borg} \\ \hline 0 & 0 & 0 & 1 \end{array}\right]$,
- Or, $\begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = \left[\begin{array}{ccc|c} & {}^{A}_{B}R & & {}^{A}P_{Borg} \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \cdot \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix}$,
- Or, ${}^{A}P = {}^{A}_{B}R \cdot {}^{B}P + {}^{A}P_{Borg}$.
- Our aim is now to solve for ${}^{B}P$ in terms of all other matrices and vectors.

Inverse of Transformation Matrix (3)

- From ${}^{A}P = {}^{A}_{B}R \cdot {}^{B}P + {}^{A}P_{Borg}$,
- We have: ${}^{B}P = {}^{A}_{B}R^{-1} \cdot ({}^{A}P - {}^{A}P_{Borg})$.
- However, we already learnt that ${}^{A}_{B}R^{-1} = {}^{A}_{B}R^{T}$.
- Therefore, ${}^{B}P = {}^{A}_{B}R^{T} \cdot ({}^{A}P - {}^{A}P_{Borg})$.
- Writing this in homogeneous transformation matrix gives:

$$\boxed{\begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix} = \left[\begin{array}{ccc|c} & {}^{A}_{B}R^{T} & & -{}^{A}_{B}R^{T} \cdot {}^{A}P_{Borg} \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \cdot \begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix}}$$

Inverse of Transformation Matrix (4)

- Summary:

- Given: $\boxed{{}^{A}_{B}T = \left[\begin{array}{ccc|c} & {}^{A}_{B}R & & {}^{A}P_{Borg} \\ \hline 0 & 0 & 0 & 1 \end{array}\right]}$,

- The inverse is: $\boxed{{}^{A}_{B}T^{-1} = {}^{B}_{A}T = \left[\begin{array}{ccc|c} & {}^{A}_{B}R^{T} & & -{}^{A}_{B}R^{T} \cdot {}^{A}P_{Borg} \\ \hline 0 & 0 & 0 & 1 \end{array}\right]}$

Inverse of Transformation Matrix (5)

- Example: {B} is rotated relative to {A} about z-axis by 30º, and then translated 4 units in X_A and 3 units in Y_A.
  - The description of {B} w.r.t. {A} is: $${}^{A}_{B}T = \left[\begin{array}{ccc|c} 0.866 & -0.5 & 0 & 4 \\ 0.5 & 0.866 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right]$$
  - And the description of {A} w.r.t. {B} is:

$${}^{B}_{A}T = \left[\begin{array}{c|c} \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}^T & -\begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}^T \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix} \\ \hline 0 \quad 0 \quad 0 & 1 \end{array}\right] = \left[\begin{array}{ccc|c} 0.866 & 0.5 & 0 & -4.964 \\ -0.5 & 0.866 & 0 & -0.598 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right]$$

    - You may verify this using Matlab.

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Background (1)

- We learnt that to represent an orientation, we could use the 3 x 3 rotation matrix:

$$ {}^{A}_{B}R = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} $$

- However, it is sometimes not easy to visualize the "projection" interpretation of the matrix.
  - E.g. ${}^{A}_{B}R = \begin{bmatrix} 0.8373 & -0.4063 & 0.3659 \\ 0.5365 & 0.7397 & -0.4063 \\ -0.1055 & 0.5365 & 0.8373 \end{bmatrix}$
  - How to sketch the coordinates? – not easy!

Background (2)

- Another issue is, there are 9 parameters in the rotation matrix.
  - Do we really need 9 parameters to describe a rotation in 3D-space?
  - It turns out that there are 6 constraints for the 9 numbers:

$$|\hat{X}| = 1, |\hat{Y}| = 1, |\hat{Z}| = 1, \hat{X} \cdot \hat{Y} = 0, \hat{X} \cdot \hat{Z} = 0, \hat{Y} \cdot \hat{Z} = 0$$

  - Therefore, it is possible to describe orientations with just 3 parameters.

Background (3)

- In this section, we will learn other interpretation of the rotation matrix which require only 3 (or 4) parameters e.g. a, b, c:

$$
{}^{A}_{B}R = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} \overset{\text{also}}{=} \begin{bmatrix} f_{11}(a,b,c) & f_{12}(a,b,c) & f_{13}(a,b,c) \\ f_{21}(a,b,c) & f_{22}(a,b,c) & f_{23}(a,b,c) \\ f_{31}(a,b,c) & f_{32}(a,b,c) & f_{33}(a,b,c) \end{bmatrix}
$$

Background (4)

- Also, if we are given a rotation matrix with numerical values:

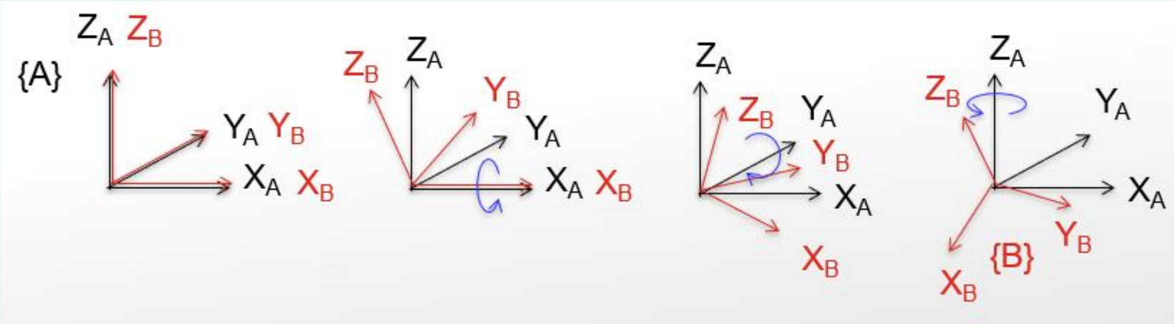
$$ {}^{A}_{B}R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}, $$

  - How do you recover the 3 (or 4) individual parameters a, b, c?

$$ {}^{A}_{B}R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} f_{11}(a,b,c) & f_{12}(a,b,c) & f_{13}(a,b,c) \\ f_{21}(a,b,c) & f_{22}(a,b,c) & f_{23}(a,b,c) \\ f_{31}(a,b,c) & f_{32}(a,b,c) & f_{33}(a,b,c) \end{bmatrix} $$

XYZ-Fixed Angles (1)

- Also known as roll, pitch, and yaw.
- To describe the orientation of {B}:
  - Start by aligning {B} with {A}.
  - First, rotate {B} about X_A by angle γ.
  - Then, rotate the rotated {B} about Y_A by angle β.
  - Finally, rotate the rotated {B} about Z_A by angle α.



XYZ-Fixed Angles (2)

- The rotation matrix is obtained by multiplying the individual rotations in the order from the right to the left:

$$
{}^{A}_{B}R = R_Z(\alpha)\cdot R_Y(\beta)\cdot R_X(\gamma)
$$

$$
= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}\cdot\begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix}\cdot\begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}
$$

$$
\boxed{= \begin{bmatrix} c\alpha\cdot c\beta & c\alpha\cdot s\beta\cdot s\gamma - s\alpha\cdot c\gamma & c\alpha\cdot s\beta\cdot c\gamma + s\alpha\cdot s\gamma \\ s\alpha\cdot c\beta & s\alpha\cdot s\beta\cdot s\gamma + c\alpha\cdot c\gamma & s\alpha\cdot s\beta\cdot c\gamma - c\alpha\cdot s\gamma \\ -s\beta & c\beta\cdot s\gamma & c\beta\cdot c\gamma \end{bmatrix}}
$$

  - where “c” is the shorthand for cosine, and “s” is the shorthand for sine.

XYZ-Fixed Angles (3)

- Now, if you are given a rotation matrix with numerical values, you can calculate the parameters α, β, γ by solving simultaneous equations:

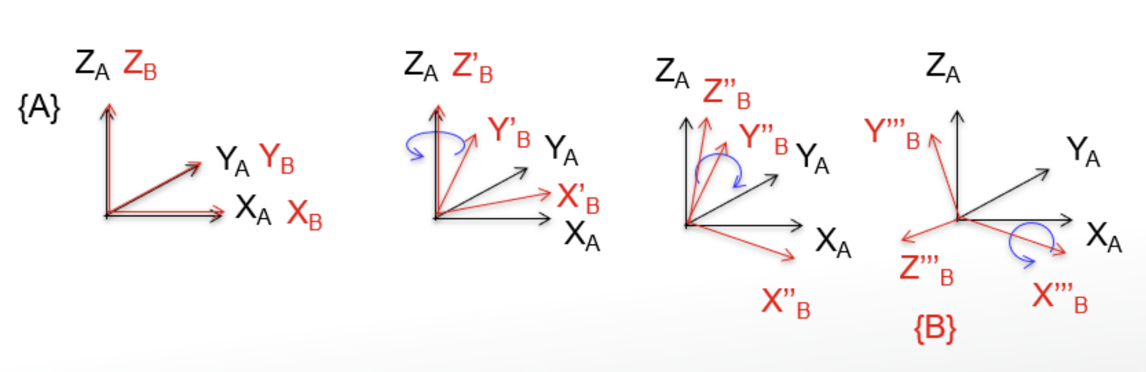
$$ {}^{A}_{B}R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix} $$

XYZ-Fixed Angles (4)

- We see that:
  - $r_{11}^2 + r_{21}^2 = c^2\beta \rightarrow c\beta = \sqrt{r_{11}^2 + r_{21}^2}$
  - $\tan\beta = \frac{s\beta}{c\beta} = \frac{-r_{31}}{\sqrt{r_{11}^2+r_{21}^2}} \rightarrow \beta = \text{artan2}\left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}\right)$
  - $\tan\alpha = \frac{s\alpha}{c\alpha} = \frac{r_{21}/c\beta}{r_{11}/c\beta} \rightarrow \alpha = \text{arctan2}\left(\frac{r_{21}}{c\beta}, \frac{r_{11}}{c\beta}\right)$
  - $\tan\gamma = \frac{s\gamma}{c\gamma} = \frac{r_{32}/c\beta}{r_{33}/c\beta} \rightarrow \gamma = \text{arctan2}\left(\frac{r_{32}}{c\beta}, \frac{r_{33}}{c\beta}\right)$
  - We can’t solve for α, γ if $c\beta = 0$.

ZYX-Euler Angles (1)

- To describe the orientation of {B}:
  - Start by aligning {B} with {A}.
  - First, rotate {B} about Z_B by angle α.
  - Then, rotate {B} about the new Y_B by angle β.
  - Finally, rotate {B} about the new X_B by angle γ.



ZYX-Euler Angles (2)

- The rotation matrix is obtained by multiplying the individual rotations in the order from the left to the right:

$$
{}_{B}^{A}R = R_Z(\alpha) \cdot R_Y(\beta) \cdot R_X(\gamma)
$$

$$
= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}
$$

$$
\boxed{= \begin{bmatrix} c\alpha \cdot c\beta & c\alpha \cdot s\beta \cdot s\gamma - s\alpha \cdot c\gamma & c\alpha \cdot s\beta \cdot c\gamma + s\alpha \cdot s\gamma \\ s\alpha \cdot c\beta & s\alpha \cdot s\beta \cdot s\gamma + c\alpha \cdot c\gamma & s\alpha \cdot s\beta \cdot c\gamma - c\alpha \cdot s\gamma \\ -s\beta & c\beta \cdot s\gamma & c\beta \cdot c\gamma \end{bmatrix}}
$$

  - The result is the same as having three rotations taken in the opposite order about fixed axis!!

ZYX-Euler Angles (3)

- Because the rotation matrix is the same, the solution to α, β, γ is also the same:
  - $\tan\beta = \frac{s\beta}{c\beta} = \frac{-r_{31}}{\sqrt{r_{11}^2+r_{21}^2}} \rightarrow \beta = \text{artan2}\left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}\right)$
  - $\tan\alpha = \frac{s\alpha}{c\alpha} = \frac{r_{21}/c\beta}{r_{11}/c\beta} \rightarrow \alpha = \text{arctan2}\left(\frac{r_{21}}{c\beta}, \frac{r_{11}}{c\beta}\right)$
  - $\tan\gamma = \frac{s\gamma}{c\gamma} = \frac{r_{32}/c\beta}{r_{33}/c\beta} \rightarrow \gamma = \text{arctan2}\left(\frac{r_{32}}{c\beta}, \frac{r_{33}}{c\beta}\right)$
  - We can’t solve for α, γ if $c\beta = 0$.

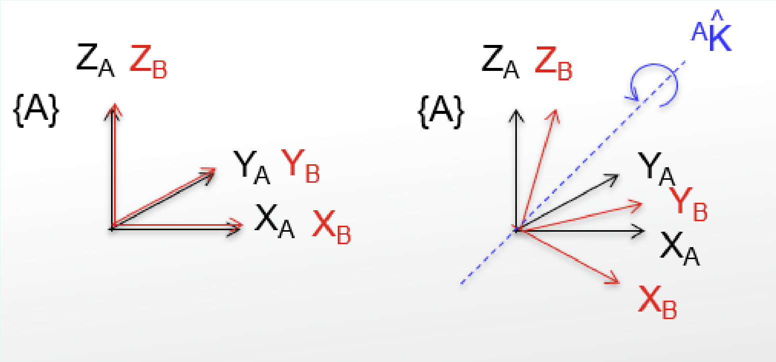
Note on Fixed Angles and Euler Angles

- The order need not be XYZ or ZYX.
- There are in fact 24 combinations!

Fixed Angles	Fixed Angles	Euler Angles	Euler Angles
X-Y-Z	X-Y-X	Z'-Y'-X'	X'-Y'-X'
X-Z-Y	X-Z-X	Y'-Z'-X'	X'-Z'-X'
Y-X-Z	Y-X-Y	Z'-X'-Y'	Y'-X'-Y'
Y-Z-X	Y-Z-Y	X'-Z'-Y'	Y'-Z'-Y'
Z-X-Y	Z-X-Z	Y'-X'-Z'	Z'-X'-Z'
Z-Y-X	Z-Y-Z	X'-Y'-Z'	Z'-Y'-Z'

Equivalent Angle Axis (1)

- Rather than 3 subsequent rotations, any rotation may also be obtained through one proper axis and angle.
- To describe the orientation of {B}:
  - Start by aligning {B} with {A}.
  - Rotate {B} about a vector ^{A}K by angle θ (according to right hand rule).



Equivalent Angle Axis (2)

- Let $\hat{K} = [k_x \quad k_y \quad k_z]^T$ be a unit vector describing the axis of rotation.
  - Since it is a unit vector, we have:

$$\boxed{k_x^2 + k_y^2 + k_z^2 = 1}$$

  - Therefore, one of the parameters is redundant, and we actually need only 2 parameters to define the vector.
- Finally, θ is the third parameter.
- Thus, the Equivalent Angle-Axis representation uses 3 parameters in total.

Equivalent Angle Axis (3)

- The equivalent rotation matrix is given by:

$$\boxed{{}^{A}_{B}R = \begin{bmatrix} k_x^2 v\theta + c\theta & k_x k_y v\theta - k_z s\theta & k_x k_z v\theta + k_y s\theta \\ k_x k_y v\theta + k_z s\theta & k_y^2 v\theta + c\theta & k_y k_z v\theta - k_x s\theta \\ k_x k_z v\theta - k_y s\theta & k_y k_z v\theta + k_x s\theta & k_z^2 v\theta + c\theta \end{bmatrix}}$$

- Where $\boxed{v\theta = 1 - c\theta}$.

Equivalent Angle Axis (4)

- Now, if you are given a rotation matrix with numerical values, you can calculate the axis $\hat{K}$ and angle θ by solving simultaneous equations:

$$
{}^{A}_{B}R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}
$$

$$
= \begin{bmatrix} k_x^2 v\theta + c\theta & k_x k_y v\theta - k_z s\theta & k_x k_z v\theta + k_y s\theta \\ k_x k_y v\theta + k_z s\theta & k_y^2 v\theta + c\theta & k_y k_z v\theta - k_x s\theta \\ k_x k_z v\theta - k_y s\theta & k_y k_z v\theta + k_x s\theta & k_z^2 v\theta + c\theta \end{bmatrix}
$$

Equivalent Angle Axis (5)

- Firstly,

$$\begin{aligned} r_{11} + r_{22} + r_{33} &= k_x^2 v\theta + c\theta + k_y^2 v\theta + c\theta + k_z^2 v\theta + c\theta \\ &= \left(k_x^2 + k_y^2 + k_z^2\right) v\theta + 3c\theta \\ &= v\theta + 3c\theta \\ &= 1 - c\theta + 3c\theta \\ &= 1 + 2c\theta \end{aligned}$$

- Therefore,

$$c\theta = \frac{r_{11}+r_{22}+r_{33}-1}{2} \text{ or } \boxed{\theta = \arccos \frac{r_{11}+r_{22}+r_{33}-1}{2}}$$

Equivalent Angle Axis (6)

- The rotation axis $\hat{K}$ can then be calculated as:

$$\boxed{\hat{K} = \frac{1}{2\sin\theta}\begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}}$$

  - Note: This solution always gives θ between 0 and 180 degrees.
  - For any $(\hat{K}, \theta)$ there is another pair, $(\widehat{-K}, -\theta)$ which gives the same rotation.
  - Also note: The solution fails if θ = 0 or θ = 180 degrees.

Equivalent Angle Axis (7)

- Example: A frame {B} which is initially coincident with {A} is rotated about the vector ${}^{A}K = [0.707\ 0.707\ 0]^T$ (passing through the origin) by 30 degrees. What is the rotation matrix from {A} to {B}?

- Answer: $v\theta = 1 - c\theta = 1 - 0.866 = 0.134.$

$$
{}^{A}_{B}R = \begin{bmatrix} k_x^2 v\theta + c\theta & k_x k_y v\theta - k_z s\theta & k_x k_z v\theta + k_y s\theta \\ k_x k_y v\theta + k_z s\theta & k_y^2 v\theta + c\theta & k_y k_z v\theta - k_x s\theta \\ k_x k_z v\theta - k_y s\theta & k_y k_z v\theta + k_x s\theta & k_z^2 v\theta + c\theta \end{bmatrix}
$$

$$
= \begin{bmatrix} 0.933 & 0.067 & 0.354 \\ 0.067 & 0.933 & -0.354 \\ -0.354 & 0.354 & 0.866 \end{bmatrix}
$$

Euler Parameters / Quaternions (1)

- We saw that in all previous methods with 3 parameters, there will be cases where no solutions can be found.
  - To tackle this issue, we need to increase to 4 parameters.
  - Euler parameters (Quaternions) is one such representation.
  - From the equivalent angle-axis representation, calculate the following Euler parameters:

$$\boxed{\varepsilon_1 = k_x \sin(\theta/2)\,,\varepsilon_2 = k_y \sin(\theta/2)\,,\varepsilon_3 = k_z \sin(\theta/2)\,,\varepsilon_4 = \cos(\theta/2)}$$

  - The parameters are not independent, and satisfy:

$$\boxed{\varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \varepsilon_4^2 = 1}$$

Euler Parameters / Quaternions (2)

- The rotation matrix is given by:

$$\boxed{{}^{A}_{B}R = \begin{bmatrix} 1-2\varepsilon_2^2-2\varepsilon_3^2 & 2(\varepsilon_1\varepsilon_2-\varepsilon_3\varepsilon_4) & 2(\varepsilon_1\varepsilon_3+\varepsilon_2\varepsilon_4) \\ 2(\varepsilon_1\varepsilon_2+\varepsilon_3\varepsilon_4) & 1-2\varepsilon_1^2-2\varepsilon_3^2 & 2(\varepsilon_2\varepsilon_3-\varepsilon_1\varepsilon_4) \\ 2(\varepsilon_1\varepsilon_3-\varepsilon_2\varepsilon_4) & 2(\varepsilon_2\varepsilon_3+\varepsilon_1\varepsilon_4) & 1-2\varepsilon_1^2-2\varepsilon_2^2 \end{bmatrix}}$$

Euler Parameters / Quaternions (3)

- Now, if you are given a rotation matrix with numerical values, you can calculate the Euler parameters by solving simultaneous equations:

$$ {}^{A}_{B}R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} 1-2\varepsilon_2^2-2\varepsilon_3^2 & 2(\varepsilon_1\varepsilon_2-\varepsilon_3\varepsilon_4) & 2(\varepsilon_1\varepsilon_3+\varepsilon_2\varepsilon_4) \\ 2(\varepsilon_1\varepsilon_2+\varepsilon_3\varepsilon_4) & 1-2\varepsilon_1^2-2\varepsilon_3^2 & 2(\varepsilon_2\varepsilon_3-\varepsilon_1\varepsilon_4) \\ 2(\varepsilon_1\varepsilon_3-\varepsilon_2\varepsilon_4) & 2(\varepsilon_2\varepsilon_3+\varepsilon_1\varepsilon_4) & 1-2\varepsilon_1^2-2\varepsilon_2^2 \end{bmatrix} $$

- Note that $r_{11}+r_{22}+r_{33}=3-4(\varepsilon_1^2+\varepsilon_2^2+\varepsilon_3^2)$.
  - Since $\varepsilon_1^2+\varepsilon_2^2+\varepsilon_3^2+\varepsilon_4^2=1$, we have $r_{11}+r_{22}+r_{33}=3-4(1-\varepsilon_4^2)$.
  - Therefore, $\boxed{\varepsilon_4=\frac{1}{2}\sqrt{1+r_{11}+r_{22}+r_{33}}.}$

Euler Parameters / Quaternions (4)

- Substituting ε_4 into ${}^A_B R$, we can get:

$$\boxed{\varepsilon_1 = \frac{r_{32} - r_{23}}{4\varepsilon_4},\ \varepsilon_2 = \frac{r_{13} - r_{31}}{4\varepsilon_4},\ \varepsilon_3 = \frac{r_{21} - r_{12}}{4\varepsilon_4}}$$

- Weird… What if $\varepsilon_4 = 0$? Didn't we say Euler Parameters will always have solutions?

Euler Parameters / Quaternions (5)

- There is a way out of this problem: We do not need ε_4 to get $\varepsilon_1, \varepsilon_2, \varepsilon_3$ as shown in previous slide.
- The alternative is:

$$\boxed{\begin{aligned} r_{11} - r_{22} - r_{33} &= -1 + 4\varepsilon_1^2 \rightarrow \varepsilon_1 = \frac{1}{2}\sqrt{1 + r_{11} - r_{22} - r_{33}} \\ -r_{11} + r_{22} - r_{33} &= -1 + 4\varepsilon_2^2 \rightarrow \varepsilon_2 = \frac{1}{2}\sqrt{1 - r_{11} + r_{22} - r_{33}} \\ -r_{11} - r_{22} + r_{33} &= -1 + 4\varepsilon_3^2 \rightarrow \varepsilon_3 = \frac{1}{2}\sqrt{1 - r_{11} - r_{22} + r_{33}} \end{aligned}}$$

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Question 1

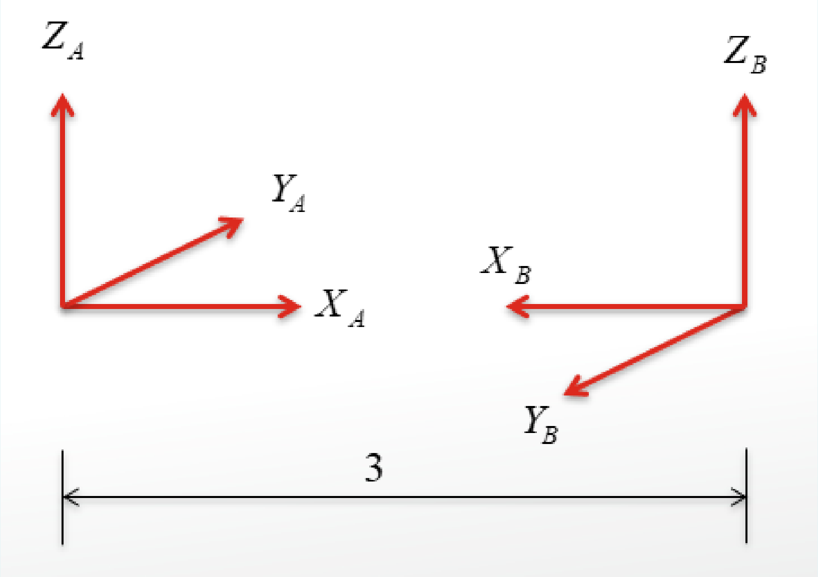
- A position vector of a point in frame {B} is given by

$$^{B}P = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix}$$

- The origin of frame {B} is at [11,-3,9] with respect to frame {A}.
- Also, frame {B} is rotated by 30 degrees along the z-axis of frame {A}.
- What is the position vector of the point in frame {A}?
- Also, write down the homogenous transformation matrix.

Question 2

- Give the value of ${}^{A}_{B}T$.



Question 3

- A vector ${}^{A}P$ is rotated about Z_A by θ degrees.
- It is subsequently rotated about X_A by Φ degrees.
- Give the rotation matrix that accomplishes these rotations in the specified order.

Question 4

- A vector ^{A}P is rotated about Y_A by 30 degrees.
- It is subsequently rotated about X_A by 45 degrees.
- Give the rotation matrix that accomplishes these rotations in the specified order.

Question 5

- A frame {B} is originally coincident with frame {A}.
- We first rotate {B} about Z_B by θ degrees.
- Then we rotate the resulting frame about X_B by Φ degrees.
- Give the rotation matrix that will change the descriptions of vectors from ${}^{B}P$ to ${}^{A}P$.

Question 6

- The axis of a particular rotation is K = [2, 1, 2]. (NOTE: this is not yet a unit vector).
- This axis passes through the origin.
- The angle of rotation is 45 degrees.
- Derive the rotation matrix based on equivalent angle-axis representation.

Question 7

- For the same rotations as in Question 6, derive the rotation matrix based on Euler parameters.

Question 8

- A rotation matrix is given by:

$$R = \begin{bmatrix} 0.8373 & -0.4063 & 0.3659 \\ 0.5365 & 0.7397 & -0.4063 \\ -0.1055 & 0.5365 & 0.8373 \end{bmatrix}$$

- Find out all the parameters in terms of:
  - X-Y-Z fixed angles
  - Z-Y-X Euler angles
  - Equivalent Angle-Axis representation
  - Euler parameters

Question 9

- The following frame definitions are given as known.

$$
{}^{U}_{A}T = \begin{bmatrix} 0.866 & -0.5 & 0 & 11 \\ 0.5 & 0.866 & 0 & -1 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^{B}_{A}T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.866 & -0.5 & 10 \\ 0 & 0.5 & 0.866 & -20 \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

$$
{}^{C}_{U}T = \begin{bmatrix} 0.866 & -0.5 & 0 & -3 \\ 0.433 & 0.75 & -0.5 & -3 \\ 0.25 & 0.433 & 0.866 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

- Solve for ${}^{B}_{C}T$.

Thank you for your attention!

Any questions?